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# Enhancing Sleep Prediction Accuracy in Lifelog Data

Using LLM-based Missing Value Imputation for Improved Sleep Quality Prediction

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# ETRI Lifelog Dataset & Research Insight Overview (1/2)

## ETRI Lifelog Dataset

**X (Input Data, 12 sensor variables)**
Multimodal smartphone and smartwatch data (charging status, activity recognition, ambient sound, BLE/Wi-Fi, GPS, illuminance, screen status, app usage, heart rate, step count, etc.) comprehensively capturing daily activities and environmental context.

**Y (Targets, 6 sleep indicators)**
Sleep health indicators derived from sensors and self-reports (Q1–Q3: subjective sleep quality, fatigue, stress; S1–S3: adherence to TST, SE, SOL guidelines), discretized into binary (0/1) or ternary (0/1/2) labels on a daily basis.

## Key Insight ( 🔥 🔥 🔥 )

Previous studies have shown that **bedtime, wake time,** and **total sleep duration** are critical variables for predicting sleep quality. However, these variables were not provided in the dataset and therefore had to be derived from the given sensor data.

Due to the nature of sensor data, noise was present, and **on certain days** the absence of sensor data made it impossible to estimate bedtime, wake time, or sleep duration.

We believe **1) handling these data quality issues** and **2) feature engineering** are the key to success – **"garbage in, garbage out"** principle applies to sleep prediction accuracy.

# ETRI Lifelog Dataset & Research Insight Overview (2/2)

## ① Dataset Summary

| | |
|---|---|
| Collection Devices | Android Smartphone & Smartwatch |
| Collection Interval | 1 to 10 minutes |
| Timestamp | Korea Standard Time (KST), YYYY-MM-DD HH:MM:SS format |
| Key Features | Contains substantial levels of missing data and noise - Cause: Device charging, system reboots, user not wearing/carrying the device, etc. |
| Privacy Protection | Anonymized by converting some data (e.g., GPS latitude/longitude) to relative coordinates |

## ② Data Items

| | | |
|---|---|---|
| Smartphone | mACStatus | Charging status (0: No, 1: Charging) |
| | mActivity | User activity type (in_vehicle, on_bicycle, walking, still, etc.) |
| | mAmbience | List of ambient sound labels and their respective probabilities |
| | mBle | List of nearby Bluetooth devices (address, device_class, RSSI) |
| | mGps | List of GPS info (altitude, latitude, longitude, speed) |
| | mLight | Ambient light in lx unit |
| | mScreenStatus | Screen usage status (0: No, 1: Using screen) |
| | mUsageStats | List of app names and their respective usage times |
| | mWifi | List of nearby Wi-Fi APs (BSSID, RSSI) |
| Smartwatch | wHr | List of heart rate recordings |
| | wLight | Ambient light in lx unit |
| | wPedo | • burned_calories: Calories burned (kcal)<br>• distance: Distance (m)<br>• speed: Speed (km/h)<br>• step: Number of steps<br>• step_frequency: Step frequency per minute |

# Bedtime and Wake Time Inference

In this competition, only sensor data were provided, information on bedtime and wake time was unavailable. Nevertheless, previous studies have demonstrated that these variables—together with their difference (i.e., sleep duration)—**are critical indicators of sleep quality and stress**.

So we decided to inference bedtime and wake time, sleep duration based on given sensor data.

## Bed time / Wakeup time inference procedure

1) **Time Window Filtering :**
Screen activity records were restricted to the range 21:00–11:00, which typically encompasses the main sleep period.

2) **Noise Reduction**
Isolated screen-on events within screen-off intervals were removed, and short awake segments (≤2 minutes) surrounded by sleep blocks were reclassified as sleep to reduce fragmentation

3) **Sleep Block Detection**
Continuous screen-off intervals were detected, and the longest interval was designated as the primary sleep episode.

4) **Feature Calculation**
The start of the longest block was defined as bedtime (21:00–02:00), the end of the block as wake time (03:00–11:00), and their difference as sleep duration (excluded if <100 minutes).

# Introduction to challenge

Despite its potential, lifelog data often suffers from missing values and noise due to the inherent limitations of real-world data collection. Even with continuous monitoring via smartphones and smartwatches, missing or anomalous entries frequently occur. This problem is exacerbated by inconsistent device usage—for instance, **users may not wear smartwatches during sleep**—resulting in incomplete or fragmented sleep-related records. Such issues limit the reliability of downstream analyses and predictive modeling.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | $\underbrace{\text{Q1} \sim \text{S3}}_{}$ Target | | |
| Bed time | - | Wakeup time | = | Sleep Duration | Q1 | ~ | S3 |
| 1:00 (Estimated) | | ??? | | ??? | 1 | | 1 |
| 2:00 (Estimated) | | 8:00 (Estimated) | | 6:00 (Estimated) | 0 | | 0 |
| ??? | | ??? | | ??? | 1 | | 1 |
| ??? | | 9:00 (Estimated) | | ??? | 0 | | 0 |

# LLM-based Imputation Framework

To address these challenges, this study proposes an imputation framework that leverages **the inference capabilities of LLMs**. By imputing missing values in sensor-derived lifelog data, we aim to enhance the informational quality of input features and thereby improve the accuracy of sleep quality prediction.

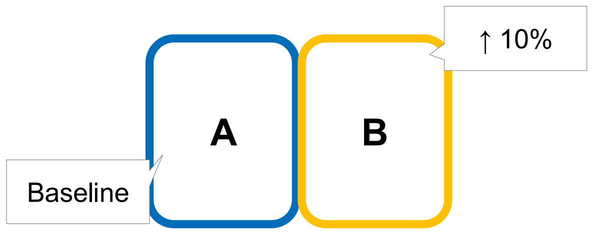
1

**Data Exploration**
Conduct exploratory data analysis to derive insights for LLM prompt engineering and context provision

*Ex)*
*Weekend effect: Later bedtimes and wake times on Fridays and Saturdays.*
*Seasonal effect: Earlier wake times during July and August.*

2

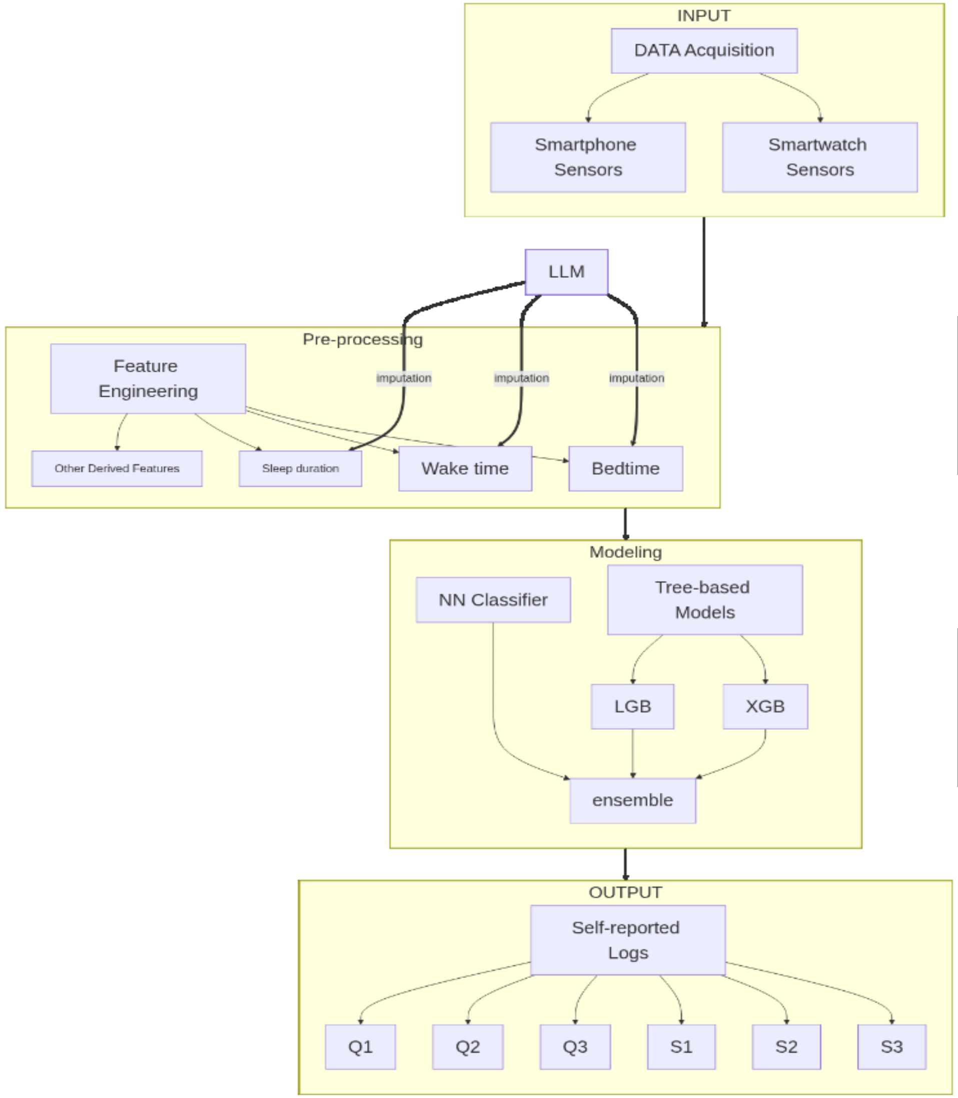
**Prompt Engineering**
Optimize prompts through A/B testing to guide LLM in contextually appropriate missing value estimation.




3

**Application**
apply optimized prompts for final missing value completion in dataset.

# Overall Model Framework




✓ LLM-based Imputation

✓ 3 Model Ensemble (TabPFN, LGB, XGB)

# Evaluation Strategy & Model Results

- ✓ Fair comparison through test-like validation set(valid size : 40) : LLM imputation **outperforms** traditional methods.
- ✓ LLM imputation **improved** performance in all targets except S3.

Comparison of F1 Scores by Different Imputation Methods

| Method | F1 | Top1 Weights (LGB, XGB, Tab) | Top1 F1 |
|---|---|---|---|
| Original | 0.6841 | (0.1, 0.2, 0.7) | 0.6998 |
| Mean Impute | 0.6684 | (0.2, 0.0, 0.8) | 0.7152 |
| KNN Impute | 0.6989 | (0.2, 0.1, 0.7) | 0.7116 |
| LLM Impute | 0.7218 | (0.3, 0.0, 0.7) | 0.7238 |

TARGET-WISE F1 SCORES

| Target | Original | Mean Impute | KNN Impute | LLM Impute |
|---|---|---|---|---|
| **Q1** | 0.716 | 0.693 | 0.697 | **0.721** |
| **Q2** | 0.780 | 0.811 | **0.840** | **0.840** |
| **Q3** | 0.715 | 0.680 | 0.715 | **0.748** |
| **S2** | 0.670 | 0.619 | 0.699 | **0.775** |
| **S3** | **0.748** | 0.723 | 0.723 | 0.723 |
| **S1** | 0.476 | 0.484 | 0.519 | **0.524** |